

ECO 641 Final Exam Project

Instructor:

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Directions:

Respond to the following discussion prompts. Your response should be clearly explained and carefully organized. Any calculations should be clearly explained and comprehensible to a reader with graduate training in economics.

Scenario:

In North Carolina, gasoline taxes are currently about \$0.60 per gallon (about \$0.20 Federal and \$0.40 State). A politician wants to eliminate the NC gasoline tax and argues that eliminating the \$0.40 tax will save North Carolina citizens \$2 billion per year based on current consumption of 5 billion gallons.

Background:

In 2019, North Carolina consumed 114,576 thousand barrels of gasoline which is about 5 billion gallons. Gasoline demand and supply are both quite inelastic. Short-run demand elasticities have been estimated around -0.2, and supply elasticities are estimated around 0.1. The retail gasoline price is approximately \$2.75 per gallon. Using these parameters, annual demand for gasoline in North Carolina is approximately given by $D(p^B) = 6 - 0.36p^B$ where demand is measured in billions of gallons and p^B is the price buyers pay in \$ per gallon. Also, annual supply to North Carolina is approximately given by $S(p^S) = 4.5 + 0.23p^S$, where supply is also in billions of gallons and p^S is the price (net of the tax) that sellers receive in \$ per gallon.

Discussion Prompts:

- 1) Using this demand and supply parameterization, how would the gasoline price change if the NC gasoline tax were eliminated? Explain your calculations.
- 2) How would consumer expenditures on gasoline change if the tax is eliminated? Do consumers save \$2 billion? Explain your calculations.
- 3) How would producer revenue from gasoline change if the tax is eliminated? Explain your calculations.
- 4) Who wins and who loses by eliminating the gasoline tax? Quantify the effects on consumers, producers, and state government revenues. Explain your calculations.
- 5) From a perspective of economic efficiency in the gasoline market, is eliminating the gasoline tax a good idea? Explain why or why not.
- 6) From a broader perspective, is eliminating the gasoline tax a good idea? Explain why or why not.